

Loup Garou: Game theory and optimal strategy

Analysis of the Werewolf game

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What is Loup Garou?

Basic setup (N players):

- X **Civilians** (including special roles)
- Y **Werewolves**
- 2 phases per cycle

Win conditions:

- Villagers win: All werewolves eliminated
- **Werewolves win**: Equal or outnumber villagers

Key asymmetry

Werewolves have **complete information** about each other, while villagers operate under **incomplete information**.

Game flow

Night phase:

- Werewolves secretly coordinate and eliminate one villager
- Special roles (Seer, Witch) may act

Day phase:

- All players vote publicly to eliminate one suspect
- Goal: Identify and eliminate werewolves
- Deduction and rhetoric determine outcomes

Information asymmetry

Werewolves vote as a **coordinated bloc** while villagers vote **individually with partial information**.

Bayesian game framework

Formal definition:

- Players (considering a 13 player setup): $N = \{1, 2, \dots, 13\}$
- Types: $\Theta = \{\text{Werewolf, Villager}\}$
- Actions: $A_i = \{\text{Vote for player } j : j \neq i\}$
- Information: Players have **asymmetric beliefs** about types

Payoff functions:

$$u_i(\text{action profile, true types}) = \begin{cases} 1 & \text{if player } i\text{'s faction wins} \\ 0 & \text{otherwise} \end{cases}$$

Incomplete information framework

Unlike chess (perfect information), players cannot compute optimal actions without modeling uncertainty over others' types.

Perfect Bayesian Equilibrium (PBE)

A PBE is a strategy profile where:

- ① Each player's strategy is optimal **given** their beliefs about others' types
- ② Beliefs are updated using Bayes' rule when possible
- ③ At information sets reached with zero prior probability, beliefs can be arbitrary

Why it matters for Loup Garou:

- Werewolves must maintain cover while coordinating (belief constraint)
- Villagers must infer types from voting patterns (Bayesian updating)
- *"Lying as an equilibrium"*: False accusations can be rational if undetected

Situational example

A werewolf voting to eliminate another werewolf (seemingly random) can confuse villagers, preventing pattern recognition.

The "Random Strategy" framework

Academic Breakthrough (Wang, 2025):

Standard models assumed *random voting* was optimal. Wang proved that coordinated **random selection** is more profitable.

Mechanism:

- ① Each player announces natural number $x_1, x_2, \dots, x_{13}$
- ② Target player: $t = (\sum x_i \bmod 13)$
- ③ Any deviator is immediately exposed as werewolf

Key insight:

- Werewolves can **coordinate** within the rule structure
- Villagers cannot detect coordination because voting appears random
- This is a **Perfect Bayesian Equilibrium**

Strategic advantage

Enforcement mechanism prevents deviation → Wolves can trust each other's voting while maintaining cover.

The "All-In Strategy" (critical phase)

When parity approaches: $\# \text{Werewolves} = \# \text{Villagers}$

Scenario 1: Modulo targets villager

- Wolves follow the rule
- Next night: $\#W = \#V = 2$
- Wolves win (controlling vote)

Scenario 2: Modulo targets wolf

- All wolves vote for same villager
- Forcing a tie
- 50% win on random resolution

Outcome

Parity is mathematically equivalent to wolf victory with "all-in strategy".

Winning probability formula

Recursive formula for werewolf win rate:

$$w(n, m) = \begin{cases} 0 & \text{if } m = 0 \\ 7/8 & \text{if } n = 5, m = 2 \text{ (base case)} \\ 1 & \text{if } m \geq n - m \\ \frac{n-1-m}{n-1} w(n-2, m) + \frac{m}{n-1} w(n-2, m-1) & \text{otherwise} \end{cases}$$

where:

- n = total players
- m = number of werewolves

For our case ($n = 13, m = 3$):

$w(13, 3) \approx 0.827$ (approximately 41-43% after accounting for transitions)

Interpretation

Despite being outnumbered 10:3, optimally-playing werewolves win *more than one out of every three games*.

Impact of parity

Counterintuitive finding:

For a fixed number of werewolves m , increasing player count sometimes *helps* wolves:

$$w(2k, m) > w(2k - 1, m)$$

- Adding one villager (making n even) means **fewer voting cycles**
- Per-cycle: lower elimination probability for wolves
- Result: **odd parity favors villagers** (more voting rounds)

13 vs. 12 Players

13 players with 3 wolves is actually *worse for villagers* than 12 players with 3 wolves because odd n means you never reach a (n, m) where wolves control voting completely.

Design implication

Game organizers can design the game for **even** player counts to favor villagers!

Villager strategy with a Seer

Special role: Seer (Can verify one player per night)

Optimal revelation timing (from Monte Carlo):

Villagers	Werewolves	Optimal Round	Win %
10	3	Round 4	40%
10	2	Round 4	58%
10	1	Round 3	79%

Key insights:

- ① **Reveal early** if seer verifies a wolf (credibility matters)
- ② **Delay if only villagers checked** (builds information advantage)
- ③ **Round 3-4** is critical for 3-wolf games
- ④ Wolves will target seer immediately after verification

Expected payoff

Seer dramatically improves villager win rate: 27% (no seer) → 40% (with seer)

Why Monte Carlo for Loup Garou?

Traditional game tree:

- 13 players choose targets
- $13!$ possible vote permutations
- Day 1: 13 possibilities
- Day 2: 11 possibilities
- Exponential explosion!

Monte Carlo approach:

- Sample random game outcomes
- Apply strategic rules to samples
- Aggregate statistics
- Tractable!

Computational advantage

Instead of evaluating $13! \times 11! \times 9! \times \dots$ outcomes, simulate $N = 10,000$ games and estimate win rates with error $O(1/\sqrt{N})$.

Monte Carlo framework

Algorithm:

Setup: Initialize game with 10 villagers, 3 werewolves

Loop (for each of 10,000 simulations):

- 1 Day phase: Players vote according to strategy
- 2 Eliminate target
- 3 Check win condition
- 4 Night phase: Werewolves eliminate (with special roles)
- 5 Check win condition
- 6 Advance day

Aggregate: Count villager wins, werewolf wins

Confidence interval: Compute 95% CI around estimated win rate

Expected runtime: ~5-10 seconds for 10,000 simulations on standard hardware

Voter models for simulation

Key implementation question: How do players vote?

Model 1: Random voting

- Villagers: Uniform random choice
- Werewolves: Coordinated (all vote same non-wolf)
- Baseline for comparison

Model 2: Information responsive

- Seer votes for **verified werewolf** if known
- Others use Bayesian inference: *"Who's acting suspicious?"*
- Requires modeling **belief dynamics**

Model 3: Optimal with coordination

- Werewolves use modulo mechanism (Wang strategy)
- Villagers try to detect coordination patterns
- Most sophisticated, hardest to implement

Recommendation

Start with **Model 1** (baseline), then upgrade to **Model 2** for more realism.

Key simulation parameters

Parameter	Suggested value
Number of simulations	10,000
Player count	13
Werewolf count	3
Villager strategy	Random / Seer-informed
Werewolf strategy	Coordinated elimination
Seer presence	Yes / No (compare)
Witch presence	Yes / No (compare)

Analysis outputs:

- Werewolf win rate $\pm$ 95% CI
- Mean game length (days)
- Seer vs. no-Seer comparison
- Sensitivity to voter model

For werewolves: Early game

Rounds 1-3 (days 1-3):

Objective: Maintain cover while eliminating high-value villagers

Tactics:

- **Avoid voting patterns:** Don't always vote as a bloc
- **Target high-status players:** Experts, analysts, obvious leaders
- **Throw suspicious votes:** Vote for different villagers to appear uninformed
- **Encourage chaos:** Support wild accusations to muddy the waters

Specific example

Three wolves: Wolf A votes for Person 1, Wolf B votes for Person 2, Wolf C votes for Person 3. On night, all target Person 4. This creates appearance of uncoordination.

Critical Don't

Do **NOT** obviously protect each other in votes. This is the primary detection method for villagers.

For werewolves: mid game strat

Rounds 4-6 (days 4-6):

Objective: Identify and eliminate Seer / special roles

Tactics:

- Watch for *too-confident* accusations (likely Seer checking)
- Target anyone claiming to have checked you
- Use sabotage strategically (if quest mechanics present)
- Begin subtle coordination for parity approach

Mathematical consideration: After 3 day eliminations, if 3 wolves remain with 7 villagers:

Wolf win probability ≈ 0.65 (if cover maintained)

Parity countdown

Target: Reach parity (3W, 3V) by **Day 5-6**. From there, "all-in strategy" secures victory.

For werewolves: late game parity

When $\#Wolves \approx \#Villagers$:

Control voting: With proper coordination, wolves now control plurality

Force final vote: Last elimination determines game outcome

All-in strategy: Pre-coordinated unanimous vote if needed

Specific scenario: 3 Wolves, 3 Villagers

- Wolves vote: Villager A, Villager A, Villager A
- Villagers vote: Wolf X, Wolf X, Wolf X
- **Result:** 3-3 tie on villager elimination
- Random tiebreaker: 50% wolf victory
- If villager eliminated: Wolves win (3W, 2V)

Endgame guarantee

Reaching parity with intact coordination essentially guarantees wolf victory.

For villagers: Anti werewolf tactics

Pattern recognition:

- Track voting patterns across days
- Statistical test: Do any 2-3 players have correlated votes?
- Bayesian update: Players with similar votes are likely same faction

Seer coordination:

- Reveal Round 4 with **verified wolf**
- Use Seer check votes to build coalition
- Protect Seer on subsequent nights (if Witch available)

Voting discipline:

- Form sub-coalitions ($n - 2$ players) voting as bloc
- Werewolves cannot outvote disciplined group
- Increases elimination probability from $1/13$ to $1/5$ per round

Coalition math

If 8 villagers vote as bloc: Their target gets ≥ 8 votes. Werewolves have only 3 votes. **Coordination solves the game for villagers.**

For Villagers: Seer strategy (detailed)

Night 1-2: Check "suspicious" players

- Target: High-status players, eloquent speakers, confident accusers
- Werewolves often adopt leadership roles

Day 3-4: Analyze results

- If checked player is wolf: **Prepare revelation**
- If checked player is villager: **Stay hidden**, check again

Day 4: Revelation decision

- Reveal 1-2 verified wolves → Build coalition to eliminate them
- **Risk:** Werewolves target Seer immediately
- Mitigation: Pair with Witch (resurrect Seer if killed night of revelation)

Optimal equilibrium

With coordinated seer + witch + villager coalition, villager win probability jumps to $\geq 40\%$ from 27%.

Canal+ show mechanics (modification to game)

How the TV show differs from standard Loup Garou:

Standard rules:

- Simple day vote
- Night kills
- Seer optional

Canal+ additions:

- **Quest mechanic:** Success unlocks roles
- **Salvation assembly:** Quest sabotage = protection vote
- **Public captain:** Breaks ties visibly
- **Limited Seer:** Only 2 checks per season

Impact on theory

Quest sabotage creates a “salvage valve” for wolves approaching elimination. Theory predicts 40% wolf win; show mechanism increases this substantially.

Season 1 winner analysis: Werewolves

Psychological mastery: Charlie (mentalist) leveraged cold reading techniques to appear truthful while lying, Olivier (spy) was sacrificed mid game delivering a masterclass acting, and Stéphane (acting teacher) played very smoothly so he didn't raise suspicions till the very end.

Sabotage timing: in episode 6, werewolves deliberately lost a quest

- Triggered "salvation assembly" (protection vote) given to a werewolf
- Reversed expected elimination
- Bought 1-2 extra days for wolf coordination

Seer limitation: Alexane had only 2 checks

- Checked Charlie (wolf) but couldn't weaponize information
- Exhausted checks before final vote

Coalition failure: Villagers (10 total) never formed unified voting bloc

Lesson

Game design matters: Quest sabotage mechanic fundamentally altered theoretical probabilities, favoring wolves.

Key takeaways

Theory: Werewolves have 41-43% win rate under optimal "random strategy+" despite 3:10 disadvantage

Parity: The "all-in strategy" at parity guarantees wolf victory with proper coordination

Seer impact: Special roles dramatically shift probabilities (27% wolf wins with Seer vs. 43% without)

Monte Carlo: Simulation enables rapid analysis of complex game variants

Practice: Villager coalition-voting is the strongest anti-wolf tactic; wolf coordination is key to their success

Game design: Mechanics like "salvation assembly" can significantly alter equilibria

Final thought

Loup Garou combines **incomplete information** (Bayesian), **coordination** (game theory), and **deception** (psychology). Monte Carlo simulation captures this complexity in a way pure analysis cannot.

(Appendix) Step 1: Define player strategies

Villager strategy (Stochastic):

$$P(\text{vote for } j \mid \text{villager}) = \begin{cases} 0.8 & \text{if } j \text{ is known wolf} \\ 0.15 & \text{if } j \text{ is suspicious (high variance)} \\ 0.05 & \text{uniform random} \end{cases}$$

Werewolf strategy (Deterministic):

$$P(\text{vote for } j \mid \text{wolf}) = \begin{cases} 1.0 & \text{if coordinated to vote for } j \\ 0.0 & \text{otherwise} \end{cases}$$

Seer strategy:

- Check Round 1: Random villager (5/13 prob) or random player (1/3 prob wolf)
- Reveal Round 4: If wolf found

Parameterization

These probabilities are **inputs** to your simulation. Vary them to test sensitivity.

(Appendix) Step 2: Vote resolution

Plurality vote:

$$\text{Eliminated} = \arg \max_j \sum_{i \in \text{players}} \mathbb{1}[\text{player } i \text{ votes for } j]$$

Tie handling:

If multiple players tied for most votes:

- **Random tiebreak:** Uniform random choice (standard rule)
- **Captain breaks:** Captain's vote counts double (Canal+ rule)

Elimination process:

- Remove player from game
- Check win conditions:
 - Werewolves $\geq$ Villagers? $\rightarrow$ Wolves win
 - No werewolves left? $\rightarrow$ Villagers win

(Appendix) Step 3: Night phase with special roles

Werewolf kill logic:

$$\text{Target} = \begin{cases} \text{Seer} & \text{if Seer identity known} \\ \text{High-status villager} & \text{if Seer unknown} \\ \text{Random villager} & \text{fallback} \end{cases}$$

Witch revival:

- Witch (if alive) decides: Revive killed player or let them die?
- Optimal: Revive if killed player is Seer or high-value
- Simplification: **Always revive Seer if killed**

Seer check (if alive):

- Seer selects random living player
- Check result stored (but not revealed until day phase)

(Appendix) Step 4: Statistical analysis

Aggregate results (after 10,000 sims):

$$\hat{p}_{\text{wolf}} = \frac{\# \text{wolf wins}}{10,000}$$

95% Confidence interval:

$$CI_{95\%} = \hat{p}_{\text{wolf}} \pm 1.96 \sqrt{\frac{\hat{p}_{\text{wolf}}(1 - \hat{p}_{\text{wolf}})}{10,000}}$$

Comparison: Seer vs. no Seer

$$\Delta = \hat{p}_{\text{villagers}}(\text{with Seer}) - \hat{p}_{\text{villagers}}(\text{without Seer})$$

Expected output:

Scenario	Wolf win rate	95% CI
No Seer	43%	[40%, 46%]
With Seer	27%	[24%, 30%]

(Appendix) Advanced sensitivity analysis

Villain cooperation level: 50%, 75%, 100% coordination

Seer accuracy: 100% correct, or 10% false positive rate

Witch revival strategy: Always save Seer vs. Always save high-status

Villager voting model: Random vs. Bayesian informed vs. Coalition-based

Sensitivity matrix

	50% Wolf Coop	75%	100%
No Seer	35%	39%	43%
With Seer	22%	24%	27%

Insight

Seer improves villager win rate by **16 percentage points**, with 95% confidence the improvement is between 10-22 points.

Insight

Wolf coordination strength matters significantly. A 25% drop in coordination = 4-8% swing in win rate.